Plan: Baseline for Pure States → Mixed States

Projective Measurements: $\hat{P}_m = |m\rangle\langle m|$, $M = \sum_m m \hat{P}_m$

e.g. $M_z = |0\rangle\langle 0| - |1\rangle\langle 1| = Z$

Probability of Result m:

$$|\langle m|\psi\rangle|^2 = \langle\psi|m\rangle\langle m|\psi\rangle = \langle\psi|\hat{P}_m|\psi\rangle = p(m)$$

Renormalization of final State: $N\hat{P}_m|\psi\rangle = |\psi'\rangle$ Set $\langle\psi'|\psi'\rangle \equiv 1$

$$1 \equiv \langle\psi'|\psi'\rangle = |N|^2 \langle\psi|P_m^\dagger P_m|\psi\rangle$$

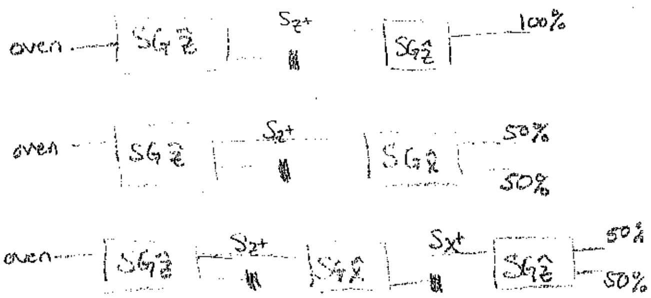
projection ops. Hermitian $P^\dagger = P$ and $P^2 = P$

$$= |N|^2 \langle\psi|P_m|\psi\rangle$$

$$= |N|^2 p(m)$$

$$\therefore N = \frac{1}{\sqrt{p(m)}} \quad \text{and} \quad |\psi'\rangle = \frac{\hat{P}_m|\psi\rangle}{\sqrt{p(m)}}$$

Physical Application:
Post Selection As Intermediate process
e.g. Stern-Gerlach Examples



i.e. not characterized simultaneously by S_z^+ and S_x^+

For M to represent A valid Measurement, "Set spans all outcomes"

$$\sum_m p(m) = 1 = \sum_m \langle\psi|\hat{P}_m|\psi\rangle = \langle\psi|\sum_m \hat{P}_m|\psi\rangle$$

$$\rightarrow \sum_m \hat{P}_m = \mathbb{1}$$

So What is M? operator for probability-weighted average observable e.g. $E(z) = (+1)P(z_+) + (-1)P(z_-)$
e.g. informs polarization

Easily Express Expectation Values: $E(M) = \sum_m m\, p(m) = \sum_m m\langle\psi|\hat{P}_m|\psi\rangle = \langle\psi|M|\psi\rangle$

Variance: $\langle (M - \langle M\rangle)^2\rangle = \langle M^2 - 2M\langle M\rangle + \langle M\rangle^2\rangle = \langle M^2\rangle - \langle M\rangle^2$

Example: Measure $|0\rangle$ in X-basis, should get 50/50 $|+\rangle, |-\rangle$

$\langle 0|X|0\rangle = 0$ i.e. unpolarized along x-axis

Standard deviation: $\sqrt{\langle X^2\rangle - \cancel{\langle X\rangle}^{2}} = \sqrt{\langle \mathbb{1}\rangle} = 1$ as all measurements are ± 1

Mixed States:

Classical Ensemble $\{p_i, |\psi_i\rangle\}$ (classical real Probabilities, $\sum_i p_i = 1$, (not Amplitudes)) $\longrightarrow \rho = \sum_i p_i \underbrace{|\psi_i\rangle\langle\psi_i|}_{\rho_{pure,i}}$

Cyclic Property of trace
$Tr(ABC) = Tr(CAB) = Tr(BCA)$
e.g. Big matrices $A_{\alpha\beta} B_{\beta\gamma} C_{\gamma\alpha}$

density Matrix Captures All observable Quantities

All probabilities Accounted for: $\sum_i p_i = 1 \Rightarrow Tr(\rho) = \sum_i p_i \, Tr\left[\,|\psi_i\rangle\langle\psi_i|\,\right] = 1$ (normalized States)

• All physical DM' have Unit trace.

• ρ is positive Semi Definite: evals are measurement probs in Eigenbasis.

• $Tr[\rho^2]$ — If ρ is pure, one element in Sum, $\rho_{pure} = |\psi\rangle\langle\psi|$

and $\rho^2_{pure} = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho_{pure}$

$\Rightarrow Tr\left[\rho^2_{pure}\right] = 1$

one non-zero eigenvalue rank 1

— If ρ is mixed, more than 1 element in Sum. >1 non-zero eigenvalue

Trace is Basis Independent
$Tr[U\rho U^\dagger] = Tr[\rho]$

evaluate in Eigenbasis:
$\sum_i p_i^2 < 1$

$\Rightarrow Tr\left[\rho^2_{mixed}\right] < 1$, "purity" (IV.A)

Use pure State + linearity to Determine Properties:

(4) Evolution: $|\psi\rangle \rightarrow U|\psi\rangle$ $\qquad \rho \rightarrow \sum_i p_i U|\psi_i\rangle\langle\psi_i|U^\dagger = U\rho U^\dagger$

why DM is not An operator' $\langle\psi|U^\dagger\rho U|\psi\rangle$ DM: state info!

(1) Projective Measurement: $p(m|i) = \langle\psi_i|\hat{P}_m|\psi_i\rangle$ $\qquad p(m|i) = Tr\left[\hat{P}_m|\psi_i\rangle\langle\psi_i|\right]$

$p(m) = \sum_i p(m|i)p(i) = Tr\left[\hat{P}_m\rho\right]$

(2) Expectation Vals: $E_i(M) = \langle\psi_i|M|\psi_i\rangle$ $\qquad E_\rho(M) = \sum_m m\,p(m) = Tr[M\rho]$

Bayes $p(m|i)p(i) = p(i|m)p(m)$

(3) After Measurement: $|\psi_i\rangle \rightarrow \dfrac{\hat{P}_m|\psi_i\rangle}{\sqrt{p(m|i)}}$ $\qquad \rho \rightarrow \rho' = \sum_i p(i|m)\dfrac{\hat{P}_m|\psi_i\rangle\langle\psi_i|\hat{P}_m^\dagger}{p(m|i)} = \sum_i p(i)\dfrac{\hat{P}_m|\psi_i\rangle\langle\psi_i|\hat{P}_m}{p(m)}$

$= \dfrac{\hat{P}_m\rho\hat{P}_m}{Tr[\hat{P}_m\rho]}$ Clearly $Tr(\rho') = 1$

Pauli Decomposition: $\rho(\vec{r}) = \dfrac{\mathbb{1} + \vec{r}\cdot\vec{\sigma}}{\underset{\substack{\uparrow \\ \text{dimension of Hilbert Space}}}{2}}$

- Generic Decomposition for Hermitian 2x2
- coefficient of $\mathbb{1}$ set to be $\frac{1}{2}$ by $Tr(\rho)=1$

for pure States,

$$\rho^2 = \rho \Rightarrow \frac{1}{4}\left(\mathbb{1} + 2\vec{r}\cdot\vec{\sigma} + \overbrace{(\vec{r}\cdot\vec{\sigma})^2}^{\vec{r}\cdot\vec{r}}\right) = \rho$$

true if $\vec{r}\cdot\vec{r} = 1$ i.e. length of $\vec{r} = 1$

lives on Unit Sphere — Sound familiar? (Ex. I.D)

e.g. $|\psi\rangle = |0\rangle$ $\rho = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ $\vec{r} = \left\{ Tr[X\rho], Tr[Y\rho], Tr[Z\rho] \right\}$

Note: Useful technique for Ex. II.D.4

$= \{0, 0, 1\}$ i.e. $\uparrow$

$|\psi\rangle = |+\rangle$ $\rho = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\frac{1}{2}$ $\vec{r} = \{1, 0, 0\}$ i.e.

mixed States:

$\rho^2 < \rho \Rightarrow \|\vec{r}\| < 1$ lives inside Bloch Sphere

maximally mixed

$$\frac{|0\rangle\langle 0|}{2} + \frac{|1\rangle\langle 1|}{2} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\frac{|+\rangle\langle +|}{2} + \frac{|-\rangle\langle -|}{2} = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Yield same observable expectation values $Tr[M\rho]$

Cannot tell by any Measure.

... Pure State Convex decompositions not unique.

Connect to Basis of lost Measurement:

Consider $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|++\rangle + |--\rangle}{\sqrt{2}}$

see from $\frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$ or insert $\mathbb{1}_X^{[4]}$

Projectively measure first Qubit

$$\rho' = \sum_k \left(\langle e_k| \otimes \mathbb{1}\right)\rho\left(|e_k\rangle \otimes \mathbb{1}\right)$$

then forget measurement = "partial tracing" = "lose access"

If measurement basis $\{|e_k\rangle\} = \{|0\rangle, |1\rangle\}$

$$\rho' = \frac{|0\rangle\langle 0|}{2} + \frac{|1\rangle\langle 1|}{2}$$

If measurement basis $\{|e_k\rangle\} = \{|+\rangle, |-\rangle\}$

$$\rho' = \frac{|+\rangle\langle +| + |-\rangle\langle -|}{2}$$

If info of first Qubit is lost to you,
could consider it A loss of measurement info in Any Basis.

$\Rightarrow$ DM captures max amount of observable information

Next time + Future Homework: Random Noise